Scalar k:2k support–two-moment cone closure

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

first issued 2026-07-28 · this version issued 2026-07-28 · v1.0

Purpose and scope

R-114 proves the exact R-112 stationary scalar $k{:}2k$ log-Laplace target for every covariance shape, every $\tau > 0$, and the closed physical cone $0 \le x \le 643\tau/200$. The gap is strict away from the origin. In particular, the complete zero-amplitude axis is closed and the R-113 directed-rounding seed box is subsumed.
The proof acts on the original uniform-phase packet. It combines a support–two-moment Hermite majorant, a self-contained centered-Bernoulli inequality, the exact R-113 phase floors, a new sharp support floor $F_b \ge -b/2$ for $b \ge 6/5$, and 3981 strictly positive exact rational Bernstein coefficients. No Bessel surrogate or floating-point quadrature enters the theorem.
This is not the all-b scalar theorem. The remaining scalar gate lies in $x > 643\tau/200$ after the inherited projective, origin, covariance-face, and phase-floor classifiers. Full A1 owner embedding, the separate one-use source/sextic aggregation, $\mathrm{OVERLAP}_{\mathrm{src}}$, Nelson, removals, the interacting measure, and Sector A remain open. R-103 REG and R-087 fixed-cutoff CORE keep their accepted owners. Historical R-085 (4.11) and (6.5) remain non-load-bearing and are not reopened.

The two executables are non-importing. The primary uses SymPy power-to- Bernstein conversion. The independent implementation uses sparse `Fraction` polynomials without SymPy and obtains selected cells by exact de Casteljau subdivision. Both reconstruct every coefficient sign, minimum, and canonical list hash. Analytic inequalities that are not finite polynomial identities are proved below.

1 Exact scalar normal form

Retain the R-112 covariance-simplex variables

$$c = \frac{v}{v+4w}, \qquad s = 1-c, \qquad b = \frac{A^2}{v+4w}, \qquad \tau = q(v+4w)^2, \qquad x = b\tau. \tag{1.1}$$

For independent unit exponentials T, U and uniform phase ϕ, put

$$\rho = \frac{cT}{2}, \qquad \sigma = \frac{sU}{8}, \tag{1.2}$$

$$L = \rho + 4\sigma - \frac{1}{2}, \quad Q = \rho^2 + 10\rho\sigma + 4\sigma^2 - \rho - \sigma, \quad B = 6\rho\sqrt{\sigma}\cos\phi. \tag{1.3}$$

The centered packet and covariance square are

$$F_b = bL + \sqrt{b}\,B + Q, \qquad \mathbb{E}F_b = 0, \tag{1.4}$$

$$K = b^2K_2 + bK_1 + K_0, \tag{1.5}$$

where

$$K_2 = c^2 - c + \frac{1}{2}, \quad K_1 = \frac{-3c^3 + 5c^2 + c + 1}{4}, \tag{1.6}$$

$$K_0 = \frac{153c^4 - 156c^3 + 82c^2 - 4c + 5}{64}. \tag{1.7}$$

Direct exponential and phase moments give

$$V := \mathrm{Var}(F_b) = \frac{K - \Delta}{2}, \qquad \Delta = \frac{33c^4 - 36c^3 + 22c^2 - 4c + 1}{64}. \tag{1.8}$$

The reserve has the useful exact minorant

$$\Delta \geq \left(\frac{c^2}{2} + \frac{(1-c)^2}{8} \right)^2 \geq \frac{1}{100}. \tag{1.9}$$

Indeed the first difference is $c^2(1-c)^2/8$, and the expression inside the square differs from $1/10$ by $5(c-1/5)^2/8$.

Writing

$$M(c, x, \tau) = \mathbb{E}e^{-\tau F_b} \quad (x = b\tau), \tag{1.10}$$

the exact target is

$$G(c, b\tau, \tau) = \frac{\tau^2 K}{4} - \log M(c, b\tau, \tau) \geq 0. \tag{1.11}$$

2 Exact support floors

The phase minimum is attained at $\cos\phi = -1$. R-113 proved the two identities

$$\begin{aligned} F_- &= (\rho + 2\sigma)^2 - \left(1 + \frac{b}{2}\right)(\rho + 2\sigma) + (5b+1)\sigma \\ &\quad + 6\rho \left(\sqrt{\sigma} - \frac{\sqrt{b}}{2} \right)^2 - \frac{b}{2}, \end{aligned} \tag{2.1}$$

$$\begin{aligned} F_- &= \rho^2 + (b/10 - 1)\rho + 4\sigma^2 + (4b-1)\sigma \\ &\quad + 10\rho \left(\sqrt{\sigma} - \frac{3\sqrt{b}}{10} \right)^2 - \frac{b}{2}. \end{aligned} \tag{2.2}$$

Completing the remaining one-variable quadratics yields the valid floors

$$F_b \geq -\beta_6(b), \qquad \beta_6 = \frac{b^2 + 12b + 4}{16}, \tag{2.3}$$

and

$$F_b \geq -\beta_{10}(b), \qquad \beta_{10} = \frac{b}{2} + \frac{(1 - b/10)_+^2}{4} + \frac{(1-4b)_+^2}{16}. \tag{2.4}$$

The polynomial branches used below are

$$\beta_{10,a} = \frac{401b^2 - 20b + 125}{400} \quad (0 \leq b \leq 1/4), \tag{2.5}$$

$$\beta_{10,b} = \frac{b^2 + 180b + 100}{400} \quad (1/4 \leq b < 10). \tag{2.6}$$

They agree at $b = 1/4$. All selected floors are positive, so every division below is legal. At the transition to the new floor,

$$\beta_{10,b}(3/2) = \frac{1489}{1600}. \tag{2.7}$$

There is a sharper floor on the large-b side. Put $y = \sqrt{\sigma}$. A direct expansion of the phase minimum gives

$$F_- + \frac{b}{2} = \rho^2 + a_b(y)\rho + y^2(4y^2 + 4b - 1), \tag{2.8}$$

$$a_b(y) = 10y^2 - 6\sqrt{b}\, y + b - 1. \tag{2.9}$$

For $b \geq 6/5$, define

$$q_b(y) = a_b(y) + 2y\sqrt{4b-1}. \tag{2.10}$$

Its discriminant as a quadratic in y is

$$12\{b + 3 - 2\sqrt{b(4b-1)}\}. \tag{2.11}$$

This is strictly negative on $b \geq 6/5$: all quantities being squared are positive, and

$$4b(4b-1) - (b+3)^2 = 15b^2 - 10b - 9 = \frac{3}{5} + \frac{(5b-6)(15b+8)}{5} > 0. \tag{2.12}$$

Thus $q_b(y) > 0$. If $a_b(y) \geq 0$, every term in (2.8) is nonnegative. If $a_b(y) < 0$, minimization over $\rho \geq 0$ gives

$$y^2(4y^2 + 4b - 1) - \frac{a_b(y)^2}{4} = \frac{(2\sqrt{d} - a_b(y))(2\sqrt{d} + a_b(y))}{4} > 0, \quad d = y^2(4y^2 + 4b - 1), \tag{2.13}$$

because $2\sqrt{d} + a_b(y) \geq q_b(y) > 0$. At $y = 0$ the right side of (2.8) is $\rho(\rho + b - 1)$. Consequently

$$\boxed{F_b \geq -\frac{b}{2} \quad (b \geq 6/5)}, \tag{2.14}$$

with equality only at $\rho = \sigma = 0$. The floor is therefore sharp (an attained support minimum), not merely a convenient estimate. The rational cut $6/5$ is a safe proof cutoff; it is not asserted to be the optimal onset of the floor. Below we use

$$\beta_\sharp(b) = \frac{b}{2} \tag{2.15}$$

on the two new slabs.

3 The support–two-moment majorant

3.1 Hermite reduction

Lemma 3.1. Let Y be centered, $Y \geq -\beta$ almost surely, $\beta > 0$, and $V = \mathbb{E}Y^2 > 0$. For $t \geq 0$, define

$$r = \frac{\beta^2 + V}{\beta}, \qquad p = \frac{\beta^2}{\beta^2 + V}. \tag{3.1}$$

Then

$$\mathbb{E}e^{-tY} \leq e^{t\beta}\{1 - p + pe^{-tr}\}. \tag{3.2}$$

Proof. Set $Z = Y + \beta \geq 0$. Let H be the quadratic Hermite interpolant of $f(z) = e^{-tz}$ at the nodes $0, r, r$. For every $z \geq 0$, the Hermite remainder is

$$f(z) - H(z) = \frac{f'''(\xi)}{3!}\, z(z-r)^2 \leq 0, \tag{3.3}$$

because $f''' < 0$. The two-point law with mass p at r and mass $1-p$ at zero has the same first two moments as Z:

$$pr = \beta, \qquad pr^2 = \beta^2 + V. \tag{3.4}$$

Since the expectation of a quadratic depends only on these moments, $\mathbb{E}H(Z) = 1-p+pe^{-tr}$. Multiplication by $e^{t\beta}$ proves (3.2). $\square$

Put

$$\Phi(t) = t\beta + \log(1-p+pe^{-tr}), \qquad p_t = \frac{pe^{-rt}}{1-p+pe^{-rt}}. \tag{3.5}$$

Then $\Phi(0) = \Phi'(0) = 0$ and

$$\Phi''(t) = r^2 p_t(1-p_t). \tag{3.6}$$

If $\beta^2 \le V$, then $p \le 1/2$ and $0 \le p_t \le p \le 1/2$. Hence

$$\Phi''(t) \le r^2 p(1-p) = V, \qquad \Phi(t) \le \frac{Vt^2}{2}. \tag{3.7}$$

3.2 A self-contained centered-Bernoulli bound

For completeness, we prove the only non-polynomial estimate used in the other case. Define the centered Bernoulli log moment-generating function by

$$g(\lambda) = \log\{(1-p)e^{p\lambda} + pe^{-(1-p)\lambda}\}.$$

It satisfies

$$g(\lambda) \le \frac{1-2p}{4\log((1-p)/p)}\lambda^2 \tag{3.8}$$

for $0 < p < 1$, with the continuous value $1/8$ at $p = 1/2$.

By exchanging the atoms and reversing λ, assume $p < 1/2$. Put $m = 1-2p$, $a = \operatorname{atanh} m$, and $u = -\lambda/2$. Direct algebra gives

$$g(\lambda) = mu + \log\cosh(u-a) - \log\cosh a. \tag{3.9}$$

Thus (3.8) is equivalent to $h(u) \ge 0$, where

$$h(u) = \frac{mu^2}{2a} - mu - \log\cosh(u-a) + \log\cosh a. \tag{3.10}$$

Writing $v = (u-a)/a$ gives

$$h'(u) = mv - \tanh(av). \tag{3.11}$$

The function $\tanh z/z$ decreases for $z > 0$, because $z\operatorname{sech}^2 z - \tanh z < 0$. Consequently h' has signs $-,+,-,+$ on the four intervals cut by $v = -1, 0, 1$. The two global minima are $u = 0$ and $u = 2a$, and direct substitution gives $h = 0$ at both. The case $p = 1/2$ follows by the limit, equivalently $\log\cosh u \le u^2/2$.

Return now to (3.5). If $\beta^2 \ge V$, set

$$y = \frac{\beta^2 - V}{\beta^2 + V} \in [0, 1). \tag{3.12}$$

Equation (3.8), after rescaling by r, gives

$$\Phi(t) \le \frac{r^2 y}{8\operatorname{atanh} y} t^2. \tag{3.13}$$

Since

$$\operatorname{atanh} y \geq y + \frac{y^3}{3}, \tag{3.14}$$

it is sufficient that

$$r^2 \leq 2K\left(1 + \frac{y^2}{3}\right). \tag{3.15}$$

The derivative of the difference in (3.14) is $y^4/(1-y^2) \geq 0$, so no external inequality is being imported.

4 Polynomial certificate and cone theorem

For any selected floor $\beta = \beta(b)$, define

$$Q_\beta = V - \beta^2, \tag{4.1}$$

$$\begin{aligned} S_\beta = {} & 6K\beta^2(\beta^2+V)^2 + 2K\beta^2(\beta^2-V)^2 \\ & - 3(\beta^2+V)^4. \end{aligned} \tag{4.2}$$

Clearing the positive denominator in (3.15) gives the exact identity

$$S_\beta = 3\beta^2(\beta^2+V)^2\left\{2K\left(1+\frac{y^2}{3}\right) - r^2\right\}. \tag{4.3}$$

At a point with $Q_\beta \geq 0$, (3.7) applies. At a point with $Q_\beta < 0$, condition $S_\beta > 0$ implies (3.15). It is therefore enough to cover each rectangle by a high-c certificate $Q_\beta > 0$ and a low-c certificate $S_\beta > 0$. The low-c region need not have $\beta^2 \geq V$ everywhere: where it does not, (3.7) already applies.

After affine transport to $[0,1]^2$, every polynomial is written in the tensor Bernstein basis. If

$$P(u,v) = \sum_{m,n} a_{mn} u^m v^n$$

has bidegree (d,e), its Bernstein coefficients are

$$\widehat{P}_{ij} = \sum_{m \leq i,\, n \leq j} a_{mn} \frac{\binom{i}{m}}{\binom{d}{m}} \frac{\binom{j}{n}}{\binom{e}{n}}. \tag{4.4}$$

Strict positivity of all coefficients proves strict positivity on the whole closed rectangle.

The exact cover is as follows.

slab	b-interval	β	S_β region	Q_β region
I_1	$[0, 1/10]$	β_6	$c \in [0, 1/2]$	$c \in [1/2, 1]$
I_2	$[1/10, 1/4]$	$\beta_{10,a}$	$c \in [0, 1/2]$	$c \in [1/2, 1]$
I_3	$[1/4, 1]$	$\beta_{10,b}$	$c \in [0, 1/2]$	$c \in [1/2, 1]$
I_4	$[1, 3/2]$	$\beta_{10,b}$	$c \in [0, 3/5]$	$c \in [3/5, 1]$
I_5	$[3/2, 2]$	$\beta_\sharp$	not needed	$c \in [0, 1]$
I_6	$[2, 643/200]$	$\beta_\sharp$	four cells	two cells

(4.5)

For I_1, I_2, I_3, each S rectangle has 289 bidegree-$(16,16)$ coefficients and each Q rectangle has 25 bidegree-$(4,4)$ coefficients. For I_4, the low-c interval is divided into the eight closed cells

$$\left[\frac{3j}{40}, \frac{3(j+1)}{40}\right], \qquad j = 0, \ldots, 7, \tag{4.6}$$

giving $8 \cdot 289 = 2312$ positive S coefficients; its high-c rectangle has 25 positive Q coefficients. This original part contributes 3279 signs.

For I_5, $Q_{\beta_\sharp}$ is positive on the four closed quarter-cells $[j/4,(j+1)/4]$, $j=0,\ldots,3$. Its bidegree is $(2,4)$, hence this adds $4\cdot 15=60$ signs. For I_6, use $Q_{\beta_\sharp}$ on

$$[0,1/10] \quad \text{and} \quad [5/8,1], \tag{4.7}$$

and $S_{\beta_\sharp}$ on the four consecutive cells

$$[1/10,37/160], \quad [37/160,19/64], \quad [19/64,29/80], \quad [29/80,5/8]. \tag{4.8}$$

The Q and S bidegrees are respectively $(2,4)$ and $(8,16)$, so I_6 adds $2\cdot 15+4\cdot 153=642$ signs. Altogether

$$3279+60+642=3981 \tag{4.9}$$

exact rational coefficient signs are positive. Both implementations also reconstruct the exact minima and canonical coefficient-list hashes recorded in their JSON artefacts. One load-bearing new minimum, on the second I_5 quarter-cell, is

$$\frac{181}{4096}>0. \tag{4.10}$$

The smallest I_6 coefficient is likewise retained exactly in the result JSON, rather than rounded in the proof contract.

Theorem 4.1 (closed physical cone). For every $c\in[0,1]$, $\tau>0$, and $0\le x\le 643\tau/200$,

$$\boxed{G(c,x,\tau)>0.} \tag{4.11}$$

At $(x,\tau)=(0,0)$, $G=0$.

Proof. Set $b=x/\tau$. The six slabs cover $0\le b\le 643/200$, including every endpoint. Lemma 3.1 and the pointwise Q/S dichotomy give

$$\log \mathbb{E}e^{-\tau F_b}\le \frac{\tau^2K}{4}. \tag{4.12}$$

In the variance case, the unused margin is

$$\frac{K}{4}-\frac{V}{2}=\frac{\Delta}{4}>0. \tag{4.13}$$

In the other case the relevant Bernstein coefficient minimum is strictly positive, so (3.15) itself is strict. Hence the gap is strict for $\tau>0$. Direct evaluation gives equality at the origin. Points with $\tau=0,x>0$ are outside this ratio chart and retain their R-113 projective owner. $\square$

5 A reusable symmetric Bessel majorant

Although Theorem 4.1 uses the original packet directly, the proof search also produced a stable future classifier:

$$\boxed{I_0(z)\le \frac{1+\cosh z}{2}.} \tag{5.1}$$

Indeed the coefficient of z^{2n} on the two sides is respectively

$$\frac{1}{4^n(n!)^2}, \qquad \frac{1}{2(2n)!}. \tag{5.2}$$

Writing $r_n=\binom{2n}{n}/4^n$, one has $r_1=1/2$ and

$$\frac{r_{n+1}}{r_n}=\frac{2n+1}{2n+2}<1. \tag{5.3}$$

Thus (5.1) is coefficientwise, with first defect $z^4/192$. In fact

$$I_0(z) \le \frac{1+\cosh z}{2} - \frac{z^4}{192}, \tag{5.4}$$

because every remaining defect coefficient is nonnegative.

The uncorrected right side is the moment-generating function of the three-point phase law with weights $1/4, 1/2, 1/4$ at $-1, 0, 1$. It preserves the physical-ray mean and variance and creates no linear origin debt. However, it changes the fixed-x projective second coefficient; one must not import the R-112 D_2 certificate into that bare surrogate. The correction in (5.4) matches through the z^4 term and is the appropriate future one-dimensional erfcx classifier. Neither surrogate is needed for, nor extends the scope of, Theorem 4.1 here.

6 Exact remaining scalar frontier

R-112 confined any unresolved point to

$$\mathcal{D}_{112} = \Big\{0 \le c \le 1,\ 0 \le x \le 8,\ 0 \le \tau \le 20,\ (2x+\tau)^2 \le 32x + 20\tau\Big\}. \tag{6.1}$$

R-114 removes the closed cone $\tau > 0$, $x \le 643\tau/200$. Consequently the live set is

$$\mathcal{D}_{112} \cap \{\tau > 0,\ x > 643\tau/200\} \setminus (\mathcal{W}_{\mathrm{proj}} \cup \mathcal{O}_{\mathrm{origin}} \cup \mathcal{F}_{\mathrm{face}} \cup \mathcal{H}_{\mathrm{phase}}), \tag{6.2}$$

with the four inherited R-113 classifiers understood exactly as defined there. A simple certified container is

$$\kappa \le c \le 1-\kappa, \qquad \theta \le \tau < \frac{1228800}{552049}, \qquad \frac{643\tau}{200} < x \le 8, \tag{6.3}$$

together with the quadratic constraint in (6.1), where

$$\kappa = \frac{1}{100552401097327200}, \qquad \theta = \frac{16}{285418875}. \tag{6.4}$$

To obtain the new upper bound, substitute $x = b\tau$ in (6.1):

$$\tau \le R(b) := \frac{32b+20}{(2b+1)^2}. \tag{6.5}$$

The function R decreases for $b \ge 0$, and

$$R(643/200) = \frac{1228800}{552049}. \tag{6.6}$$

Both coordinate axes are therefore closed: $x = 0, \tau > 0$ by Theorem 4.1, the origin by equality, and $\tau = 0, x > 0$ by the inherited projective theorem. The R-113 Arb seed is also subsumed, since its maximal ratio is $(101/100)/(99/100) = 101/99 < 643/200$. It remains valid independent evidence but is no longer load-bearing.

7 Method boundary and proof roadmap

The cubic atanh proxy is not a global all-b argument with the selected support floor. Only $3/800$ beyond the certified endpoint, at the exact point

$$(b,c) = \left(\frac{103}{32}, \frac{5}{16}\right), \quad \beta = \frac{103}{64}, \quad K = \frac{18714321}{4194304}, \quad V = \frac{145635}{65536}, \tag{7.1}$$

one has

$$Q_\beta = -\frac{24109}{65536} < 0, \tag{7.2}$$

and

$$S_\beta = -\frac{127544381197984065}{18446744073709551616} < 0. \tag{7.3}$$

This is an exact failure of this sufficient polynomial proxy, not a counterexample to the exact log-Laplace target, the full Kearns–Saul coefficient, a sharper support floor, or a higher-moment majorant. It also does not prove optimality of the rational endpoint 643/200.

The other proof-search branches are retained with their present status:

- Exact integration of the base e^{-T-U} weight makes the R-113 two-dimensional Bessel cubature substantially tighter at fixed cell count. It remains a sound fallback, but outer parameter dependency still prevents a finite global cover; no exploratory endpoint is theorem evidence here.
- Exact fourth-order Taylor moment patches show prototype margins, while a four-moment Gauss–Radau Hermite majorant repairs nearby post-cone tests. Neither has a replayable global cover; Gauss–Radau still needs a uniform all-tilt certificate, since blanket variance monotonicity is false. It is the preferred analytic target, with exact intervals as fallback.
- The symmetric majorant (5.1) permits exact elimination of the T integral and a one-dimensional erfcx integral in U. Keeping the three branches as one mixture preserves their cancellation; bounding them separately loses it. Outer splitting is still required.
- On the residual, the coordinates $r = x + \tau$ and $p = x/(x + \tau)$ turn the quadratic exterior constraint into $r(1 + p)^2 \leq 20 + 12p$ and place the new cone boundary at $p = 643/843$. They reduce repeated-variable inflation and are preferred for the next branch-and-bound.

The next fail-closed proof route is therefore:

1. use Theorem 4.1 as the first classifier and discard every box with $x \leq 643\tau/200$;
2. retain the R-113 projective, origin, face, and high-q classifiers;
3. first prove or refute the four-moment Gauss–Radau all-tilt step, then try the corrected symmetric Bessel classifier and exact low-order moment/Bernstein patches on the strict residual;
4. use weighted exact-Bessel outward-rounded integration only as the fallback;
5. accept scalar closure only after every residual leaf has a finite analytic or strict directed-rounding certificate.

Only after that global scalar gate closes may the proof return to the full A1 one-fresh-root contraction-closed owner cluster and then to the separate one-use source/sextic Carleson aggregation.

8 Devil's-advocate review

1. **Objection:** The Hermite remainder has the wrong sign outside the interpolation interval. **Verdict: DISMISSED.** For every $z \geq 0$, including $z > r$, the factor $z(z - r)^2$ is nonnegative and $f'''(\xi) < 0$. Thus the quadratic is a global majorant on the entire support.
2. **Objection:** The low-c slabs silently assume $\beta^2 \geq V$. **Verdict: VALID WITH EXPLICIT MITIGATION.** That sign does change inside the slabs. The proof uses a pointwise dichotomy: $Q_\beta \geq 0$ invokes (3.7), while only $Q_\beta < 0$ invokes the globally positive S_β certificate. No uniform sign assumption is made.

3. **Objection:** A decimal scan was promoted to a continuum proof. **Verdict: DISMISSED.** All 3981 signs are exact rational Bernstein coefficients. The independent program reconstructs them with a different sparse-polynomial engine and de Casteljau subdivision. Sampling is absent from the evidence contract.

4. **Objection:** The Bessel surrogate supplies the missing theorem but changes the projective coefficient. **Verdict: DISMISSED.** Theorem 4.1 applies Lemma 3.1 directly to the original uniform-phase packet. Section 5 is explicitly non-load-bearing and records the fixed-x caveat.

5. **Objection:** The equality and chart boundaries were omitted. **Verdict: DISMISSED.** Every b and c slab is closed and overlaps at its endpoints, $b = 643/200$ is included, $\tau > 0$ gives strict gap, and the origin is checked directly. The separate $\tau = 0, x > 0$ boundary retains its R-113 owner.

6. **Objection:** The negative witness (7.3) refutes the scalar target. **Verdict: UPHELD AS A FORBIDDEN READING.** It refutes only the displayed cubic sufficient proxy with the selected floor. The exact target remains numerically unrefuted and mathematically open for part of $b > 643/200$.

7. **Objection:** Closing both scalar axes closes Sector A. **Verdict: UPHELD AS A FORBIDDEN READING.** The strict residual $b > 643/200$, full A1 owner embedding, one-use source/sextic aggregation, $\text{OVERLAP}_{\text{src}}$, Nelson, removals, and the interacting measure remain open. Tier stays T4.

8. **Objection:** The rational cutoff 6/5 or the cone endpoint 643/200 was proved optimal. **Verdict: VALID WITH EXPLICIT MITIGATION.** They are convenient certified bounds. The nearby witness blocks only the chosen cubic proxy; no optimal onset, maximal cone, or target failure is asserted.

9 Result footer

```
Result ID: A13-CLASSII-SCALAR-K2K-SUPPORT-TWO-MOMENT-CONE-CLOSURE
Precise statement: R-114 proves the original R-112 scalar target strictly
  for every c in [0,1], tau>0, and 0<=x<=643*tau/200.  The origin is equality.
  Exact support/Hermite/Bernoulli analysis plus 3981 positive rational
  Bernstein coefficients provide the certificate.
Scope: Stationary scalar k:2k model and the stated physical cone only; not
  the remaining b>643/200 scalar core or a full A1 theorem.
Dependencies: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION;
  R-087; R-103; R-105; R-112; R-113.
Evidence grade: ANALYTIC, EXACT, EXECUTED.
Reproduction command:
  E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/
  a13_classii_scalar_k2k_support_two_moment_cone_verify.py
Expected output: Integrated R-114 PASS; both non-importing children PASS;
  each reconstructs 3981/3981 positive exact Bernstein coefficients.
Falsification gate: Any failed moment, floor, Hermite/Kearns--Saul identity,
  coefficient sign/minimum/hash, child run, source/PDF hash, or a claim that
  the residual b>643/200 compact core is closed.
Tier before / after: T4 / T4.
No-overclaim statement: No mixed all-b scalar theorem, full A1 owner cluster,
  one-use source/sextic aggregation, OVERLAP_src, Nelson bound, removal
  limit, interacting measure, Sector-A closure, or higher tier is established.
Next required action: Run the fail-closed classifier/branch-and-bound only on
  b>643/200; after scalar closure, build the full A1 owner cluster and the
  separate one-use aggregation.
```